

■■ How to Run RMI Arithmetic Program

Java Distributed Systems — Practical Exam Guide

STEP 0 — Keep All 4 Files in ONE Folder

All files must be in the same folder. Example:

```
ArithmeticService.java  ArithmeticServiceImpl.java  ArithmeticServer.java  
ArithmeticClient.java
```

■■ Files must NOT have (1) in their names. Rename if needed.

STEP 1 — Open Command Prompt (CMD)

Press **Win + R** → type **cmd** → press Enter

Navigate to your folder:

```
cd "C:\Users\Srushti Nitin Pagar\Desktop\PPA Home\DS"
```

STEP 2 — Compile All Files

Run this command (compiles all 4 files at once):

```
javac *.java
```

■ No output = SUCCESS. You will see 4 new .class files created.

■ If error says 'class should be in file named...' → rename files, remove (1) from names.

STEP 3 — Open TWO CMD Windows

You need 2 separate CMD windows open at the same time:

How to open 2nd CMD window	
Method 1 (Easiest)	Press Win + R → type cmd → Enter (do this TWICE)
Method 2	Right-click CMD in taskbar → click "Command Prompt"
Method 3	In File Explorer → click address bar → type cmd → Enter (do twice)

In BOTH windows, navigate to your folder:

```
cd "C:\Users\Srushti Nitin Pagar\Desktop\PPA Home\DS"
```

STEP 4 — Run the SERVER (CMD Window 1) — Run FIRST!

Option A — Standard way:

```
java ArithmeticServer
```

Option B — With classpath (use this if Option A gives ClassNotFoundException):

```
java -cp . ArithmeticServer
```

■ You should see: Arithmetic Server is ready...

■ DO NOT close this window! Server must keep running while client is used.

■ 'Port already in use: 1099' → Server is already running (that's fine! Don't run it again).

STEP 5 — Run the CLIENT (CMD Window 2) — Run AFTER Server!

Option A — Standard way:

```
java ArithmeticClient
```

Option B — With classpath (use this if Option A gives ClassNotFoundException):

```
java -cp . ArithmeticClient
```

■ You should see: Enter the first number:

STEP 6 — Interact with the Program

```
Enter the first number: 10 Enter the second number: 5 Choose an
operation: 1. Add 2. Subtract 3. Multiply 4. Divide Enter your choice
(1-4): 1 Result: 15.0
```

Quick Reference — All Commands

Command	What it does
<code>javac *.java</code>	Compile all Java files → creates .class files
<code>java ArithmeticServer</code>	Start server (standard)
<code>java -cp . ArithmeticServer</code>	Start server (with classpath — use if standard fails)
<code>java ArithmeticClient</code>	Start client (standard)
<code>java -cp . ArithmeticClient</code>	Start client (with classpath — use if standard fails)
<code>dir *.class</code>	Check if .class files were created
<code>dir *.java</code>	Check your java file names
<code>Ctrl + C</code>	Stop the server

Common Errors & Fixes

Error	Fix
ClassNotFoundException	Use: <code>java -cp . ArithmeticClient</code>
Port already in use: 1099	Server already running — skip to running client
Connection refused	Start SERVER first, then client
class should be declared in file named...	Rename file — remove (1) from filename
Cannot divide by zero	Expected! Enter non-zero for divisor

■ Remember for Exam: Compile → Server first → Client second → Both terminals open together